

Functional Specification

Status legend: **Implemented** · **Partial** · **Stub** / **Planned**

1. Authentication & session

Capability	Status	Notes
Staff login (username/password)	Implemented	JWT returned
Current user profile (/auth/me)	Implemented	
Change password	Implemented	Min 8 chars; confirm match
Remember-me (local vs session storage)	Implemented	Frontend
Failed-attempt logout	Implemented	LoginAttemptService
Role-based route guards	Implemented	Angular authGuard

2. Dashboard

Capability	Status	Notes
Summary counts (customers, accounts, employees)	Implemented	
Today's transaction count	Partial	Hardcoded 0 until Transaction module exists

3. Customers

Capability	Status	Notes
Search / paginate customers	Implemented	
View customer detail	Implemented	
Create customer	Implemented	Optional first account on create
Update customer	Implemented	
Soft/hard delete	Implemented	ADMIN only for DELETE
Open additional accounts from detail	Implemented	Current (policy-aware opening deposit); Loan button present

4. Accounts

Capability	Status	Notes
Open account	Implemented	Policy-validated opening deposit
List / search accounts	Implemented	/app/accounts
List accounts by customer	Implemented	
Get account by id	Implemented	GET /accounts/{id}
Update account status	Implemented	API present
Deposit (post-open)	Implemented	Credits balances + writes CREDIT ledger row
Withdraw / transfer	Stub	Not built
Account detail + ledger UI	Stub	Not built
Account policies (create + get active)	Implemented	Seeded INR policies; no LOAN seed
Opening-deposit dialog (Current)	Implemented	Uses GET /account-policies

5. Employees (users)

Capability	Status	Notes
List / create / update employees	Implemented	ADMIN
Role assignment	Implemented	Seeded roles include unused TELLER / AUDITOR / CUSTOMER

6. Branch

Capability	Status	Notes
Branch master data	Stub	Only branchCode string (often 0001)

7. Loans

Capability	Status	Notes
Loan product / EMI / collateral	Stub	
AccountType.LOAN account opening	Partial	Enum + UI affordance; needs policy; not a loan domain

8. Transactions & reports

Capability	Status	Notes
Transaction ledger entity/API	Partial	Write on deposit + GET <code>.../transactions</code> list implemented; no withdraw/transfer/UI; dashboard count stub
Reports	Stub	Empty report package; sidebar placeholder
Beneficiaries	Stub	Empty beneficiary package

9. Audit

Capability	Status	Notes
JPA created/updated by fields	Partial	<code>AuditableEntity</code> on some entities
Audit event API / UI	Stub	Empty audit package

Non-functional (current)

Area	Behavior
AuthN	JWT HMAC, 1h expiry (<code>jwt.expiration</code>)
AuthZ	Role matchers in <code>SecurityConfig</code>
CORS	localhost, bankone.local, 192.168.0.4
Logging	Hibernate SQL DEBUG (dev-oriented)
Deploy	Open Liberty WAR primary path

Deferred hardening items are tracked in `.cursor/rules/deferred-hardening.mdc`.